Critical one-half energy thresholds and paired curvature concentration

23 September 2026. Continuation of the compact-regime realization theorem. The increment is at the scale-invariant exponent: supercritical control or critical equiintegrability is replaced by a fixed small local critical energy threshold. The small chirality may vary from ball to ball.

1. Conventions and statements

Use the Hilbert–Schmidt curvature-operator convention

$$|W^{\pm}|^2 = \tfrac{1}{4} W^{\pm}_{ijkl} (W^{\pm})^{ijkl}. \tag{1}$$

This is not the largest-eigenvalue norm. In the full tensor convention, all energy thresholds below must be multiplied by four. Let $\omega_4 = \pi^2/2$, and let $V_{-K/3}(R)$ be the volume of a radius-R ball in the simply connected four-dimensional space form of sectional curvature $-K/3$. For fixed $K, D, v > 0$, set

$$\theta_0 = \frac{v}{V_{-K/3}(D+1)}, \qquad S_0 = \frac{9}{16\sqrt{\omega_4 \theta_0}}, \qquad \kappa_0 = \frac{1}{6S_0^2} = \frac{128\omega_4 \theta_0}{243}, \qquad \epsilon_0 = \frac{\kappa_0}{4}. \tag{2}$$

If the class below is nonempty then $0 < \theta_0 \leqslant 1$. These are sufficient constants, not claimed to be sharp.

The topological assumption is

$$\pi_1(M) \text{ torsion-free}, \qquad \pi_2(M) = 0. \tag{H}$$

Every closed aspherical manifold has (H). The topological core-scope extension shows that half-flat ALE core exclusion needs only (H): the universal cover has $H_1 = H_2 = 0$, and a finite-fundamental-group core lifts to that cover. In particular, everything below may also be read in the older aspherical scope.

Theorem A (critical local compactness). Fix $K, D, v, r_0 > 0$ and $0 < \alpha < 1$. There is a uniform positive lower bound for the $C^{1,\alpha}$ harmonic radius on closed connected smooth oriented four-manifolds satisfying (H),

$$\| \mathrm{Ric}^{\sharp} \|_{\infty,\mathrm{op}} \leqslant K, \qquad \mathrm{diam}(M,g) \leqslant D, \qquad \mathrm{Vol}(M,g) \geqslant v, \tag{3}$$

and

$$\sup_{x \in M} \min \left\{ \int_{B_{r_0}(x)} |W^+|^2 \, d\mu, \ \int_{B_{r_0}(x)} |W^-|^2 \, d\mu \right\} \leqslant \epsilon_0. \tag{4}$$

Consequently the class has geometric $C^{1,\beta}$ and weak $W^{2,p}$ precompactness, and only finitely many smooth types. Here $\beta < 1$ and $p < \infty$. No uniform L^q, $q > 2$, bound or vanishing equiintegrability modulus is required.

Theorem B (critical-regime realization). Fix $D, v, r_0 > 0$. Use (2) with $K = 2$. There is $\delta_0(D, v, r_0) > 0$ such that a closed connected smooth oriented four-manifold satisfying (H) and carrying a metric with these diameter/volume bounds, (4), and

$$\delta_-(g) = \| \mathrm{Ric}^{\sharp}_g + I \|_{\infty,\mathrm{op}} < \delta_0 \tag{5}$$

admits a smooth metric h with $\mathrm{Ric}_h = -h$.

The simpler condition $\min\{\int_M |W^+|^2, \int_M |W^-|^2\} \leqslant \epsilon_0$ implies (4), but is usually unnecessarily restrictive. Condition (4) allows substantial total Weyl energy and different choices of the small half in different regions.

2. A Sobolev–Weitzenböck gap on the exact blowup

Lemma C. Let X be a complete noncompact smooth Ricci-flat oriented four-manifold, ALE of some positive order, with asymptotic volume ratio $\theta \geqslant \theta_0 > 0$. Each Weyl half separately satisfies

$$W^\pm \equiv 0 \quad \text{or} \quad \int_X |W^\pm|^2 \, d\mu \geqslant \kappa_0. \tag{6}$$

This is a classical analytic mechanism, not a classification of general Ricci-flat ALE spaces.

Proof. Brendle's nonnegative-Ricci Sobolev inequality applied to compactly supported $f = |u|^3$, by approximation at zeros, gives

$$4(\omega_4\theta)^{1/4}\left(\int |u|^4\right)^{3/4} \leqslant 3\int |u|^2|\nabla u| \leqslant 3\left(\int |u|^4\right)^{1/2}\left(\int |\nabla u|^2\right)^{1/2}.$$

Consequently

$$\|u\|_4^2 \leqslant S(\theta)\|\nabla u\|_2^2, \qquad S(\theta) = \frac{9}{16\sqrt{\omega_4\theta}} \leqslant S_0. \tag{7}$$

For either half let $w = |W^\pm|$. On the Ricci-flat limit the Einstein Weitzenböck identity reads

$$\tfrac{1}{2}\Delta w^2 = |\nabla W^\pm|^2 - 18\det W^\pm, \qquad \Delta = \operatorname{div}\nabla. \tag{8}$$

For a real symmetric trace-free 3×3 matrix,

$$|\det W^\pm| \leqslant \frac{w^3}{3\sqrt{6}}. \tag{9}$$

Indeed, with eigenvalues summing to zero,

$$(\textstyle\sum \lambda_i^2)^3 - 54(\lambda_1\lambda_2\lambda_3)^2 = 2\displaystyle\prod_{i<j}(\lambda_i - \lambda_j)^2 \geqslant 0.$$

ALE asymptotics of a positive order τ suffice for exhaustion: $W = O(r^{-2-\tau})$, $\nabla W = O(r^{-3-\tau})$. Einstein elliptic regularity gives the derivative asymptotics. Thus $w \in L^2 \cap L^4$, $\nabla W \in L^2$, and boundary errors in (8) vanish. Cutoffs and Kato extend (7) to w. No assertion about the optimal ALE coordinate order is used. Integration followed by Hölder, Kato and (7) yields

$$\begin{aligned} J := \int |\nabla W^\pm|^2 &= 18\int \det W^\pm \leqslant \sqrt{6}\int w^3 \\ &\leqslant \sqrt{6}\,\|w\|_2\|w\|_4^2 \leqslant \sqrt{6}\,S_0\|w\|_2 J. \end{aligned} \tag{10}$$

If $\|w\|_2^2 < \kappa_0$, the last coefficient is less than one, so $J = 0$. A parallel L^2 tensor on this infinite-volume connected manifold vanishes. This proves (6). □

The Sobolev estimate is on an **exact Ricci-flat blowup** with a controlled volume ratio. It does not add a hidden uniform Sobolev assumption to the earlier unrestricted monopole integral-gap theorem.

3. Topology forces both halves to concentrate together

Suppose a sequence satisfying (H) and (3) has minimum harmonic radius $\rho_j \to 0$, attained at x_j. The harmonic-radius argument of the preceding continuation gives a complete smooth nonflat Ricci-flat limit

$$(M_j, \rho_j^{-2} g_j, x_j) \longrightarrow (X, g_\infty, x_\infty). \tag{11}$$

Here are the necessary inputs, rather than an implicit invocation of smooth almost-Einstein compactness.

Relative Bishop comparison gives, at every centre and for $r \leqslant D+1$,

$$\operatorname{Vol} B_r(x) \geqslant \frac{v V_{-K/3}(r)}{V_{-K/3}(D+1)} \geqslant \theta_0 \omega_4 r^4. \tag{12}$$

Every fixed-radius ball in the rescaled limit obeys this Euclidean lower bound, so its asymptotic volume ratio is at least θ_0. Ricci tends uniformly to zero after rescaling. The normalized harmonic-radius lower bound gives C^1 charts; their elliptic metric equation gives strong $W^{2,p}$ convergence for every finite p, then a smooth Ricci-flat limit. Strong $C^{1,\alpha}$ recovery makes the limiting harmonic-radius normalization nontrivial, excluding a flat limit. No derivative of the original Ricci error is estimated.

Jiang–Naber's local full L^2 curvature estimate, summed over the bounded global covering provided by (3), bounds the scale-invariant full energy. The bound passes to (11). Finite energy and Euclidean growth give ALE structure by the Ricci-flat asymptotic theorem. Only a positive ALE order is needed here.

The geometric convergence embeds every large smooth compact core of X in M_j. If $W^+_\infty = 0$, the half-flat compact-core exclusion contradicts that embedding. If $W^-_\infty = 0$, the same argument applies with reversed orientation. Therefore both halves of this **same** blowup are nonzero, and Lemma C gives

$$\int_X |W^+|^2 \geqslant \kappa_0, \qquad \int_X |W^-|^2 \geqslant \kappa_0. \tag{13}$$

For every fixed $r > 0$, strong local curvature convergence and exhaustion in the rescaled radius imply

$$\liminf_j \int_{B_r(x_j)} |W^+_{g_j}|^2 \geqslant \kappa_0, \qquad \liminf_j \int_{B_r(x_j)} |W^-_{g_j}|^2 \geqslant \kappa_0. \tag{14}$$

To see this, a fixed radius-R ball in (11) is eventually contained in the unscaled radius-r ball. First let $j \to \infty$, then $R \to \infty$.

Paired-concentration alternative. In the bounded Ricci, diameter and noncollapse regime (3), harmonic-radius degeneration on the topological class (H) forces a definite critical amount of **both** Weyl energies at a common shrinking location, in the precise sense (14), after a subsequence.

This is stronger than saying that one half must concentrate. It does not assert the existence of non-half-flat Ricci-flat ALE spaces or assume their classification. It identifies the curvature required of any surviving bubble. If such bubbles are excluded by additional arguments, compactness strengthens further.

4. Proofs of compactness and realization

Proof of Theorem A. If the harmonic radii were not uniformly positive, apply (14) with $r = r_0$. Both energies at x_j eventually exceed $\kappa_0/2$, contradicting (4) and $\epsilon_0 = \kappa_0/4$. No globally consistent choice of chirality in (4) is needed. The controlled harmonic atlases give the stated compactness and smooth-type finiteness exactly as in the preceding continuation. □

Proof of Theorem B. Failure would give input metrics with $\delta_-(g_j) \to 0$, the fixed bounds and (4), but no normalized negative Einstein metric on their smooth manifolds. For large j the Ricci bound (3) holds with $K = 2$. Theorem A gives a subsequence identified with a fixed smooth manifold, and harmonic-coordinate $C^{1,\alpha}$ convergence. Pass to

$$g^{ab}\partial_a\partial_b g_{ij} = -2\,\mathrm{Ric}_{ij} + Q_{ij}(g, \partial g).$$

The limiting metric weakly solves $\mathrm{Ric} = -g$. Elliptic bootstrapping makes it smooth. Transporting this metric through the smooth identifications supplies the forbidden Einstein metric on each late original manifold. □

An actual monopole projection budget $(a_g^+)^2 \geqslant A_0 > 0$ can replace the lower volume hypothesis: if $\delta_- \leqslant 1/2$, the scalar estimate gives $V \geqslant 8\pi^2 A_0/9$. Substitute this v into (2). This observation does not delete actual solvability or square costs from other theorems.

5. Increment and the remaining original-problem gap

The preceding L^q, $q > 2$, hypothesis and uniform critical equiintegrability each imply (4) at a common sufficiently small radius. Conversely, (4) does not itself impose uniform vanishing of small-ball energy: it permits a fixed nonzero subthreshold amount. No actual new geometric family is claimed merely from this measure-theoretic weakening.

The variable-chirality form is useful when global control of one fixed half is unavailable but the two energies have different spatial distributions. It provides a mathematically precise potential role for chiral information in existence, rather than just another global curvature coefficient.

For a hypothetical negative-side Yau26 counterexample sequence in (H), with bounded diameter and a uniform positive actual projection budget, harmonic radii must degenerate along a subsequence; otherwise the smooth-type realization argument produces Einstein metrics. Consequently (14) holds at suitable centres: the remaining compact-regime escape must spend critical energy in **both** halves at the same location.

Diameter escape, collapse without a positive volume budget, and paired concentration above threshold remain possible. No dimension-only Yau26 threshold is proved. Sobolev and Weyl gap estimates are classical ingredients; the increment is their explicit composition with the topological core exclusion to obtain critical, locally variable-chirality compactness and same-type realization. No exhaustive publication-priority claim is made.

6. Sources and finite verification

- S. Brendle, *Sobolev inequalities in manifolds with nonnegative curvature*, Theorem 1.1: https://arxiv.org/html/2009.13717. Only the exact blowup uses this theorem. Equation (7) is the displayed elementary deduction.
- A. Derdziński, *Self-dual Kähler manifolds and Einstein manifolds of dimension four*: https://www.numdam.org/article/CM_1983___49_3_405_0.pdf. The harmonic-half-Weyl identity,

with convention (1), is explicitly recorded in Section 3 of Gursky–McKeown–Tyrrell, *Self-dual and even Poincaré–Einstein metrics in dimension four*: https://arxiv.org/html/2407.08851. Only the local identity is used, not their Poincaré–Einstein boundary hypotheses.

- W. Jiang and A. Naber, *L2 Curvature Bounds on Manifolds with Bounded Ricci Curvature*, Theorem 1.6: https://arxiv.org/html/1605.05583. Its separate derivative theorem has additional hypotheses and is not used.
- Bando–Kasue–Nakajima, *On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth*: https://doi.org/10.1007/BF01389045. Only positive-order ALE asymptotics of the finite-energy Ricci-flat limit are needed.
- E. P. Wright, *Quotients of gravitational instantons*, Theorem A, Remark 1 and Corollary C: https://arxiv.org/html/1102.2442. Its classification covers the half-flat case. Remark 1 explicitly separates the conjectural general Ricci-flat ALE classification, which is not assumed here.
- Inherited compact-core proof and topological scope extension.

The accompanying exact checker verifies the determinant polynomial, threshold arithmetic, absorption margin and critical scaling. It does not certify the compactness, asymptotic or topological arguments.